Q: Find the co-ordinates of $\begin{pmatrix}1\\2\\3\end{pmatrix}$ in $\left\{\begin{pmatrix}1\\-1\\0\end{pmatrix}, \begin{pmatrix}0\\1\\-1\end{pmatrix}, \begin{pmatrix}1\\0\\-1\end{pmatrix}\right\}$?

~~(1,2,3)~~ e_1, e_2, e_3

$S \subseteq V$ $\{b_1, b_2, \dots\}$ finite elements $\rightarrow$ finite-dimensional vector space

infinite element

($n \times n$ matrices)

$\dim(V) = \#$ basis vectors

e.g. $\mathbb{R}^3$ $\dim(\mathbb{R}^3) = 3$

$\dim(V) = 4$

Part-B

Linear Transformation:

Def^n: Let V and W be v.s over $\mathbb{F}$. A linear transformation from V to W is a function T from V to W such that

$$T(\alpha x + y) = \alpha T(x) + T(y) \quad \forall \quad x, y \in V, \ \alpha \in \mathbb{F}$$

$$T : V \longrightarrow W$$

e.g. V = space of n-degree polynomials in x.

$D : V \longrightarrow V$ $\qquad p_1 = x^3 + x^2 + 1$

$D(p) := \frac{d}{dx} p \quad \forall \ p \in V$ $\qquad D(p_1) = 3x^2 + 2x$

$$D(\alpha p_1 + p_2) = \frac{d}{dx}(\alpha p_1 + p_2) = \alpha \frac{dp_1}{dx} + \frac{dp_2}{dx} = \alpha D(p_1) + D(p_2)$$

$T : V \longrightarrow V$

$T(p) := p + 1$ $\qquad T(x+1) = x + 2$

$$T(\alpha p_1 + p_2) = \alpha p_1 + p_2 + 1$$

$$\alpha T(p_1) + T(p_2) = \alpha(p_1 + 1) + p_2 + 1 \neq T(\alpha p_1 + p_2)$$

Def^n: Nullspace Let $T : V \longrightarrow W$

$$\text{null}(T) := \{x \in V \mid T(x) = 0\}$$

Range

$$\text{range}(\tau) = \{ y \in \mathcal{W} \mid \exists x \in \mathcal{V} \quad \tau(x) = y \}$$

(nullspace & range a vector space?)

Linear transformation $\longleftrightarrow$ matrices

$$\tau : \mathcal{V} \longrightarrow \mathcal{W}$$

nullspace τ $\quad \tau(x) = 0$

$\dim(\ker(T)) = \dim(\text{nullspace } \tau)$

range τ

$\dim(\text{rng } T) = \dim(\text{rng } \tau)$

$\mathcal{B}_{\mathcal{V}} = \{ v_1, v_2, \ldots, v_n \}$ $\quad \dim(\mathcal{V}) = n$

$\mathcal{B}_{\mathcal{W}} = \{ w_1, w_2, \ldots, w_m \}$ $\quad \dim(\mathcal{W}) = m$

$\tau(v_1) \in \mathcal{W}$

$$\tau(v_1) = \alpha_{11} w_1 + \alpha_{21} w_2 + \cdots + \alpha_{m1} w_m \Rightarrow [w_1 \; w_2 \cdots w_m] \begin{bmatrix} \alpha_{11} \\ \alpha_{21} \\ \vdots \\ \alpha_{m1} \end{bmatrix} \quad (\alpha_{11}, \alpha_{21}, \ldots \alpha_{m1})$$

$$\tau(v_2) = \alpha_{12} w_1 + \alpha_{22} w_2 + \cdots + \alpha_{m2} w_m \Rightarrow [w_1 \; w_2 \cdots w_m] \begin{bmatrix} \alpha_{12} \\ \alpha_{22} \\ \vdots \\ \alpha_{m2} \end{bmatrix} \quad (\alpha_{12}, \alpha_{22}, \ldots \alpha_{m2})$$

$$\vdots$$

$$\tau(v_n) = \alpha_{1n} w_1 + \alpha_{2n} w_2 + \cdots + \alpha_{mn} w_m$$

$$\tau(x) = \tau(\beta_1 v_1 + \beta_2 v_2 + \cdots \beta_n v_n)$$

$$= \tau(\beta_1 v_1) + \tau(\beta_2 v_2) + \cdots + \tau(\beta_n v_n)$$

$$= \beta_1 \tau(v_1) + \beta_2 \tau(v_2) + \cdots + \beta_n \tau(v_n)$$

$$= [w_1 \; w_2 \; \cdots \; w_m] \underbrace{\begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & & & \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{bmatrix}}_{T \quad m \times n} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}$$

Co-ordinates in $\mathcal{W}$ after transformation of x $\tau(x)$

Linear transformation

τ $\quad T \quad \mathcal{V} \quad \mathcal{W}$

$\mathcal{B}_{\mathcal{V}} \quad \mathcal{B}_{\mathcal{W}}$

e.g. $\tau : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$

$$\tau\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\mathcal{B}_{\mathbb{R}^3} = \{ e_1, e_2, e_3 \}$ $\quad \mathcal{B}_{\mathbb{R}^2} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$$\mathcal{B}_{\mathbb{R}^3} = \left\{ \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix} \right\} \qquad \mathcal{B}_{\mathbb{R}^2} = \left\{ \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix} \right\}$$

$$\mathcal{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$$

$$\left.\begin{aligned}
\mathcal{T}(e_1) &= \begin{pmatrix}1\\0\end{pmatrix} = 1\begin{pmatrix}1\\0\end{pmatrix} + 0\begin{pmatrix}0\\1\end{pmatrix} \\
\mathcal{T}(e_2) &= \begin{pmatrix}0\\1\end{pmatrix} = 0\begin{pmatrix}1\\0\end{pmatrix} + 1\begin{pmatrix}0\\1\end{pmatrix} \\
\mathcal{T}(e_3) &= \begin{pmatrix}0\\0\end{pmatrix} = 0\begin{pmatrix}1\\0\end{pmatrix} + 0\begin{pmatrix}0\\1\end{pmatrix}
\end{aligned}\right\} \quad T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathcal{T}\begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix} \qquad T\begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}$$

$$\mathcal{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$$

$$\mathcal{B}_1 = \left\{ \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix} \right\} \qquad \mathcal{B}_2 = \left\{ \begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix} \right\}$$

$$\mathcal{T}(e_1) = \begin{bmatrix}1\\0\end{bmatrix} = a_{11}\begin{bmatrix}1\\1\end{bmatrix} + a_{21}\begin{bmatrix}1\\-1\end{bmatrix} \Rightarrow \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\begin{bmatrix}a_{11}\\a_{21}\end{bmatrix} \Rightarrow \begin{bmatrix}a_{11}\\a_{21}\end{bmatrix} = \begin{bmatrix}1/2\\1/2\end{bmatrix}$$

$$\mathcal{T}(e_2) = \begin{bmatrix}0\\1\end{bmatrix} = a_{12}\begin{bmatrix}1\\1\end{bmatrix} + a_{22}\begin{bmatrix}1\\-1\end{bmatrix} \Rightarrow \begin{bmatrix}a_{12}\\a_{22}\end{bmatrix} = \begin{bmatrix}1/2\\-1/2\end{bmatrix}$$

$$\mathcal{T}(e_3) = \begin{bmatrix}0\\0\end{bmatrix} = \overset{a_{13}}{0}\begin{bmatrix}1\\1\end{bmatrix} + \overset{a_{23}}{0}\begin{bmatrix}1\\-1\end{bmatrix}$$

$$T = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \end{bmatrix}$$

$$x = \begin{pmatrix}1\\2\\3\end{pmatrix} \qquad \mathcal{T}\begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\begin{bmatrix}\alpha\\\beta\end{bmatrix} \Rightarrow \begin{bmatrix}\alpha\\\beta\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}^{-1}\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}3/2\\-1/2\end{bmatrix}$$

$$\begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}3/2\\-1/2\end{bmatrix}$$

Commutative diagram.

$$\begin{tikzcd}
\mathcal{V} \quad \mathcal{B}^1_{\mathcal{V}} \ \mathcal{V} \arrow[r, "\overset{m\times n}{T}"] & \mathcal{W} \ \mathcal{B}^1_{\mathcal{W}} \\
\mathcal{B}^2_{\mathcal{V}} \ \mathcal{V} \arrow[u, "P"] \arrow[r, "\hat{T}"] & \mathcal{W} \ \mathcal{B}^2_{\mathcal{W}} \arrow[u, "W"']
\end{tikzcd}$$

dim n, dim m

Basis change

$$\overset{m\times n}{T}\,\overset{n\times n}{P} = \overset{m\times m}{W}\,\overset{m\times n}{\hat{T}}$$

$$\Rightarrow \underline{\underline{W^{-1}}}TP = \underline{\underline{\hat{T}}}$$

e.g.

$$\begin{tikzcd}
* \ (\alpha_1, \alpha_2, \alpha_3) \ \mathcal{B}^1_{\mathbb{R}^3} = \{e_1, e_2, e_3\} \ \mathbb{R}^3 \arrow[r, "T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}"] \arrow[d, "P^{-1}"'] & \mathbb{R}^2 \ \mathcal{B}^1_{\mathbb{R}^2} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \\
\mathcal{B}^2_{\mathbb{R}^3} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\} \ \mathbb{R}^3 \arrow[u, "P"] \arrow[r, "\hat{T}"] & \mathbb{R}^2 \ \mathcal{B}^2_{\mathbb{R}^2} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\} \arrow[u, "W = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}"']
\end{tikzcd}$$

$* \ (\gamma_1, \gamma_2, \gamma_3)$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \underline{\underline{\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}}\begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{bmatrix} \Rightarrow \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_{P}\begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{bmatrix}$$

$$TP = W\hat{T} \Rightarrow \hat{T} = W^{-1}PT$$

$$\Rightarrow \hat{T} = \begin{bmatrix} 1 & 1/2 & 1/2 \\ 0 & -1/2 & 1/2 \end{bmatrix}$$

e.g.

$$\begin{tikzcd}
* \ \mathbb{R}^3 \arrow[r, "A"] & \mathbb{R}^3 \\
\mathbb{R}^3 \arrow[u, "T"] \arrow[r, "\hat{A}"] & \mathbb{R}^3 \arrow[u, "T"']
\end{tikzcd}$$

$\mathcal{B}_1 = \{e_1, e_2, e_3\}$

$T \leftarrow$ controllability form | cont^{r} canonical form

$$\mathcal{B}_2 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$AT = T\hat{A} \Rightarrow \boxed{\hat{A} = T^{-1}AT}$$

$\{w_1, w_2 \dots w_r\} \dots$

$\mathcal{V} = \boxed{\text{img } \mathcal{C}_{AB}}$

$\mathcal{V} \subseteq \mathbb{R}^n$

$$\begin{tikzcd} \mathbb{R}^n \arrow[r, "A"] & \mathbb{R}^n \\ \mathbb{R}^n \arrow[u, "T"] \arrow[r, "\hat{A}"] & \mathbb{R}^n \arrow[u, "T"'] \end{tikzcd}$$

$(e_1, e_2 \dots e_n)$

$* \mathcal{B}_1 =$

$\boxed{\dot{x} = Ax + Bu}$

$\mathcal{V} \subseteq \mathbb{R}^n$ n vectors

$\mathcal{V} = \langle w_1, w_2 \dots w_r \rangle$

$\dim(\mathcal{V}) = r$

$\dim(\mathbb{R}^n) = n$

$(n-r)$

$\{\underbrace{w_1, w_2 \dots w_r}_{\mathbb{R}^n} \overbrace{v_1 \dots v_{n-r}}^{\text{arbitrary}}\}$

$\mathcal{B}_2 = \{w_1, w_2 \dots w_r, v_1, \dots v_{n-r}\}$

$$T = [w_1 \;\; w_2 \;\; \dots \;\; w_r \;\; v_1 \;\; \dots \;\; v_{n-r}]$$

$$AT = T\hat{A}$$

$$\Rightarrow A[\underbrace{w_1 \;\; w_2 \;\; \dots \;\; w_r}_{A\text{-invariant}} \;\; v_1 \;\; v_2 \;\; \dots \;\; v_{n-r}] = [\underbrace{w_1 \;\; w_2 \;\; \dots \;\; w_r}\;\; v_1 \;\; \dots \;\; v_{n-r}]\hat{A}$$

$$= [\qquad\qquad] \begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{bmatrix}$$

$$T = [b \;\; Ab \;\; \dots \;\; A^{n-1}b]$$

$$A[b \;\; Ab \;\; \dots \;\; A^{n-1}b] = [b \;\; Ab \;\; \dots \;\; A^{n-1}b] \underbrace{\begin{bmatrix} 0 & 0 & \dots & -a_0 \\ 1 & 0 & & -a_1 \\ 0 & 1 & & \vdots \\ \vdots & 0 & & -a_{n-1} \end{bmatrix}}_{\hat{A}}$$

$$\begin{tikzcd} & \mathbb{R}^n \arrow[r, "A"] & \mathbb{R}^n \\ u \arrow[ur, "B"] \arrow[dr, "\hat{B}"'] & & \\ & \mathbb{R}^n \arrow[uu, "T"] \arrow[r, "\hat{A}"] & \mathbb{R}^n \arrow[uu, "T"'] \end{tikzcd}$$

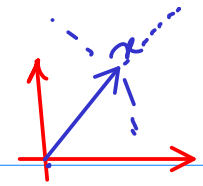
$$\boxed{T\hat{B} = B \Rightarrow \hat{B} = T^{-1}B}$$

$$\boxed{\begin{aligned} &\dim(\text{range } \mathcal{T}) = \text{rank } T = \dim(\text{img } T) \\ &\dim(\text{nullspace } \mathcal{T}) = \dim(\ker T) \end{aligned}}$$

$\mathcal{T} : \mathcal{V} \longrightarrow \mathcal{W}$

Rank-nullity thm:- $\quad \text{rank } T + \text{nullity}(T) = \dim(\mathcal{V})$

Eigenvalues & eigenvectors: $A \in \mathbb{R}^{n\times n}$



$$Ax = \lambda x$$

$\lambda \in \sigma(A)$ $\qquad \underline{\underline{(A-\lambda I)x = 0}}$ $\qquad x \in \ker(A-\lambda I)$

$\det(sI-A)$ number of times λ gets repeated as a root $\underline{\underline{n_\lambda^a}}$

is algebraic multiplicity

e.g $\quad A = \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 2 \end{array}\right]$ $\qquad \underline{\underline{n_1^a = 2}}$ $\qquad n_2^a = 1$

Geometric multiplicity: $\dim(\ker(A-\lambda I)) = n_\lambda^g$

$\lambda = 1$ $\qquad A - \lambda I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\qquad \ker(A-I) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $\qquad \underline{\underline{n_1^g = 1}}$

$A - 2I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\qquad \ker(A-2I) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $\qquad n_2^g = 1$

$(n_\lambda^a \geq n_\lambda^g)$

If $n_\lambda^a = n_\lambda^g \Rightarrow \exists\, T$ s.t. $T^{-1}AT$ is diagonal

$n_\lambda^a > n_\lambda^g \Rightarrow$ nondiagonizable

$$\begin{bmatrix} 1 & & \\ & 2 & \\ & & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & \\ & 2 & \\ & & 3 \end{bmatrix} \qquad \begin{bmatrix} \boxed{\begin{matrix} 1 & 1 \\ 0 & 1 \end{matrix}} & \\ & \end{bmatrix}$$

$$\boxed{\begin{matrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{matrix}} \qquad \boxed{\begin{matrix} \lambda & 1 \\ 0 & \lambda \end{matrix}} \qquad \boxed{\begin{matrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{matrix}}$$

$\{3, 2\}$

e.g. $\quad n_1^a = 3 \qquad n_2^a = 2$

$\quad n_1^g = 2 \qquad n_2^g = 2$

$\Downarrow$

$\dim(\ker(A-I))$

$$A = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 0 & 0 & & \\ 0 & 3 & 1 & & \\ 0 & 0 & 3 & & \\ & & & 2 & 0 \\ & & & 0 & 2 \end{bmatrix}$$